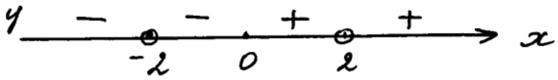
Практическое занятие 13

Построение графиков функции

1) $y = \frac{x}{(x^2-4)^2}$, $D(y) = (-\infty;-2) \cup (-2;2) \cup (2;+\infty)$



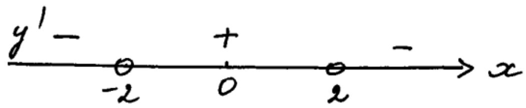
$-y(-x) = y(x)$

$\lim\limits_{x\to-2} y = -\infty,\ \lim\limits_{x\to2} y = +\infty$

$x = 2,\qquad x = -2$ − вертикальные асимптоты

$\lim\limits_{x\to\infty} y = 0, y = 0$ − горизонтальная асимптота

$$y' = \frac{(x^2-4)^2 - 2(x^2-4)2x * x}{(x^2-4)^4} = \frac{x^2-4-4x^2}{(x^2-4)^3} = -\frac{3x^2+4}{(x^2-4)^3}$$

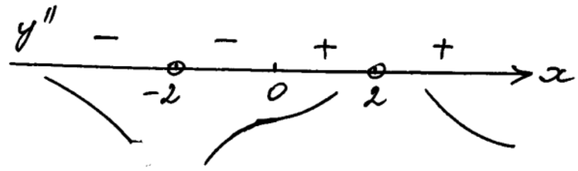


Экстремумов нет

$$y'' = -\frac{6x(x^2-4)^3 - 3(x^2-4)^2 2x(3x^2+4)}{(x^2-4)^6}$$

$$= -\frac{6x(x^2-4) - 6x(3x^2+4)}{(x^2-4)^4} = \frac{6x(3x^2+4-x^2+4)}{(x^2-4)^4}$$

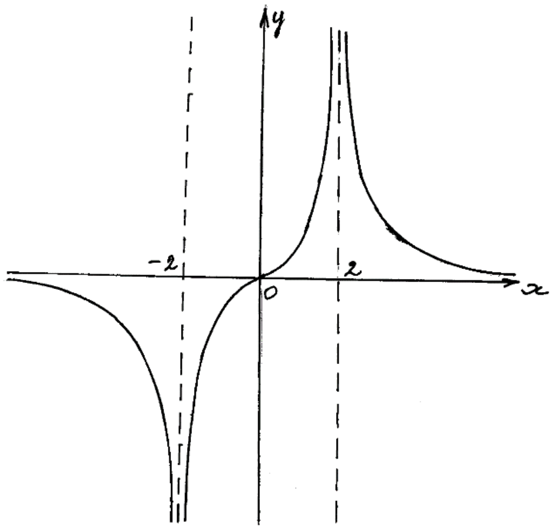
$$= \frac{12x(x^2+4)}{(x^2-4)^4}$$



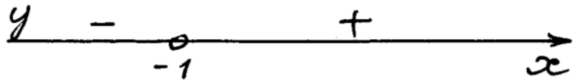
$x = 0$ − точка перегиба

$y(0) = 0$

x	$(-\infty;-2)$	-2	(-2;0)	0	(0;2)	2	$(2;+\infty)$
y’	-	∄	+		+	∄	-
y’’	-	∄	-	0	+	∄	+
y		x=-2(в а)		$y_{пер}$=0		x=2(в а)	



2) $y = \frac{x^4}{(x+1)^3}$, $\qquad D(y) = (-\infty; -1) \cup (-1; +\infty)$



$\lim\limits_{x \to -1-0} y = -\infty,$

$\lim\limits_{x \to -1+0} y = +\infty,$

x = -1 – вертикальная асимптота

$$к = \lim_{x \to \infty} \frac{y}{x} = \lim_{x \to \infty} \frac{x^3}{(x+1)^3} = \lim_{x \to \infty} \left(\frac{x}{x+1}\right)^3 = 1$$

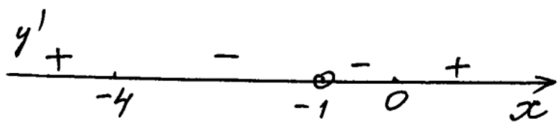
$$в = \lim_{x \to \infty} (y - к\, x) = \lim_{x \to \infty} \left(\frac{x^4}{(x+1)^3} - x\right) = (\infty - \infty)$$

$$= \lim_{x \to \infty} \frac{x^4 - x(x^3 + 3x^2 + 3x + 1)}{(x+1)^3} = \lim_{x \to \infty} \frac{-3x^3 - 3x^2 - x}{x^3 + 3x^2 + 3x + 1}$$

$$= -3$$

$y = x - 3$ – наклонная асимптота

$$y' = \frac{4x^3(x+1)^3 - 3(x+1)^2 x^4}{(x+1)^6} = \frac{4x^3(x+1) - 3x^4}{(x+1)^4} = \frac{x^4 + 4x^3}{(x+1)^4}$$

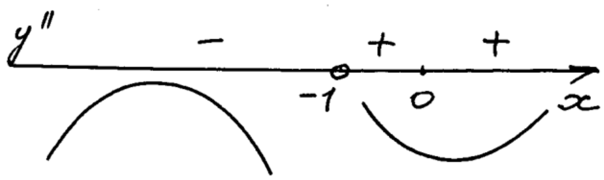


$$= \frac{x^3(x+4)}{(x+1)^4}$$

$x = -4$ – точка максимума, $\qquad y(-4) = \frac{4^4}{(-3)^3} = -\frac{256}{27}$

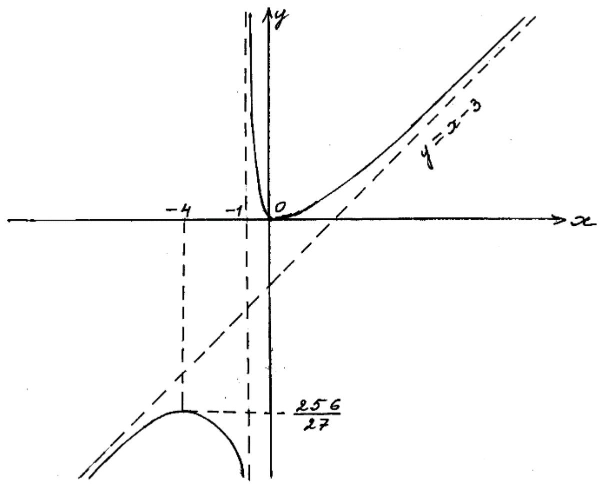
$x = 0$ – точка минимума, $\qquad y(0) = 0$

$$y'' = \frac{(4x^3 + 12x^2)(x + 1)^4 - 4(x + 1)^3(x^4 + 4x^3)}{(x + 1)^8}$$
$$= \frac{(4x^3 + 12x^2)(x + 1) - 4(x^4 + 4x^3)}{(x + 1)^5}$$
$$= \frac{4x^4 + 12x^3 + 4x^3 + 12x^2 - 4x^4 - 16x^3}{(x + 1)^5} = \frac{12x^2}{(x + 1)^5}$$

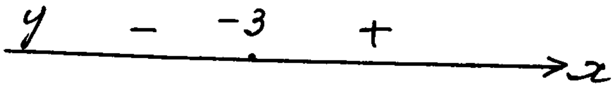


Перегибов нет

x	$(-\infty; -4)$	-4	(-4;-1)	-1	(-1;0)	0	$(0; +\infty)$
y’	+	0	-	∄	-	0	+
y’’	-		-	∄	+		+
y		$y_{max}=-\frac{256}{27}$		x=-1(в а)		$y_{min} = 0$	



3) $y = \frac{x+3}{\sqrt{x^2+1}}$ $D(y) = R$



Вертикальных асимптот нет.

$$к = \lim_{x\to\infty} \frac{y}{x} = \lim_{x\to\infty} \frac{x + 3}{x\sqrt{x^2 + 1}} = \lim_{x\to\infty} \frac{1 + \frac{3}{x}}{\sqrt{x^2 + 1}} = 0$$

$$в = \lim_{x\to\infty} (y - кx) = \lim_{x\to\infty} \frac{x + 3}{\sqrt{x^2 + 1}}$$

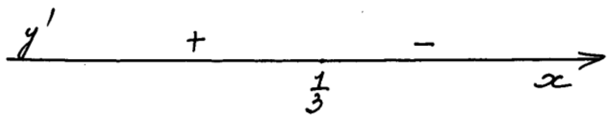
$$в = \lim_{x \to +\infty} \frac{1 + \frac{3}{x}}{\sqrt{1 + \frac{1}{x^2}}} = 1$$

$y = 1$ — правая гоизонтальная асимтота

$$в = \lim_{x \to -\infty} \frac{1 + \frac{3}{x}}{-\sqrt{1 + \frac{1}{x^2}}} = -1$$

$y = -1$ — левая гоизонтальная асимтота

$$y' = \frac{\sqrt{x^2+1} - (x+3)\frac{x}{\sqrt{x^2+1}}}{x^2+1} = \frac{x^2 + 1 - x^2 - 3x}{(x^2+1)^{\frac{3}{2}}} = \frac{-3x+1}{(x^2+1)^{\frac{3}{2}}}$$

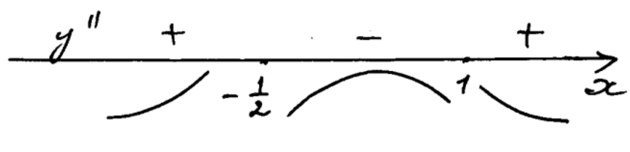


$x = \frac{1}{3}$ — точка максимума, $\quad y\left(\frac{1}{3}\right) = \frac{\frac{1}{3} + 3}{\sqrt{\frac{1}{9} + 1}} = \sqrt{10}$

$$y'' = \frac{-3(x^2+1)^{\frac{3}{2}} - (-3x+1)\frac{3}{2}(x^2+1)^{\frac{1}{2}}2x}{(x^2+1)^3} = \frac{-3(x^2+1) + 3x(3x-1)}{(x^2+1)^{\frac{5}{2}}}$$

$$= \frac{3(2x^2 - x - 1)}{(x^2+1)^{\frac{5}{2}}} = \frac{3(x-1)(2x+1)}{(x^2+1)^{\frac{5}{2}}}$$

$x = -\frac{1}{2}, \quad x = 1$ — точки пергиба

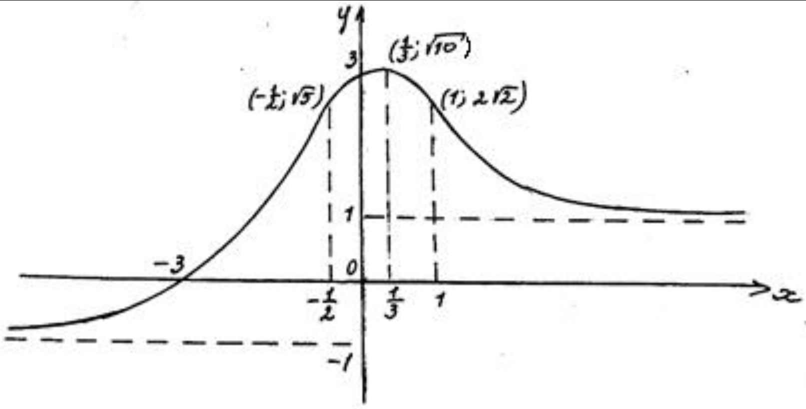


$$y\left(-\frac{1}{2}\right) = \frac{-\frac{1}{2}+3}{\sqrt{\frac{1}{4}+1}} = \sqrt{5}$$

$$y(1) = \frac{4}{\sqrt{2}} = 2\sqrt{2}$$

x	$(-\infty; -\frac{1}{2})$	$-\frac{1}{2}$	$(-\frac{1}{2}; \frac{1}{3})$	$\frac{1}{3}$	$(\frac{1}{3}; 1)$	1	$(1; +\infty)$

y'	+		+	0	-		-
y''	+	0	-		-	0	+
y		$y_{пер}=\sqrt{5}$		$y_{max} = \sqrt{10}$		$y_{пер} = 2\sqrt{2}$	



4) $y = \sqrt[3]{(x+4)^2} + \sqrt[3]{(x-4)^2}$, $D(y) = R$, $y > 0$, $\forall x \in R$

$$y(-x) = \sqrt[3]{(-x+4)^2} + \sqrt[3]{(-x-4)^2} = \sqrt[3]{(x-4)^2} + \sqrt{(x+4)^2} = y(x)$$

$y(x)$ – четная функция

Вертикальных асимптот нет.

$$к = \lim_{x\to\infty} \frac{y}{x} = \lim_{x\to\infty} \frac{\sqrt[3]{(x+4)^2} + \sqrt[3]{(x-4)^2}}{x}$$

$$= \lim_{x\to\infty} \left(\sqrt[3]{\frac{(x+4)^2}{x^3}} + \sqrt[3]{\frac{(x-4)^2}{x^3}} \right) = 0$$

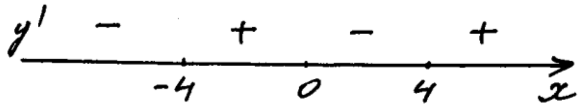
$$в = \lim_{x\to\infty} (y - кx) = \lim_{x\to\infty} y = \infty$$

Наклонных асимптот нет.

$$y' = \frac{2}{3\sqrt[3]{x+4}} + \frac{2}{3\sqrt[3]{x-4}} = \frac{2}{3} \frac{\sqrt[3]{x-4} + \sqrt[3]{x+4}}{\sqrt[3]{(x+4)(x-4)}}$$

$y'(x) = 0$: $\sqrt[3]{x-4} + \sqrt[3]{x+4} = 0 \quad \sqrt[3]{x-4} = -\sqrt[3]{x+4}$

$x - 4 = -x - 4, \qquad x = 0$



$y'(x) = \infty$: $x = -4$, $x = 4$

$$y''(x) = -\frac{2}{9\sqrt[3]{(x+4)^4}} - \frac{2}{9\sqrt[3]{(x-4)^4}} < 0 \text{ при } \forall x \in R \quad x \neq \pm 4$$

x	*0*	*(0;4)*	*4*	*(4;+ ∞)*
y'	*0*	-	∞	+
y''		-	∞	-

y	$y_{max} = 4\sqrt[3]{2}$		$y_{min} = 4$